Questions

Q1.

When white light crosses the boundary between air and glass, it can split up into the colours of the spectrum.

Explain, in terms of speed, why the light behaves like this.

(3)

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(Total for question = 3 marks)

Q2.

Sonar is an example of a use of ultrasound.

Figure 1 shows the depth of the sea, measured using sonar, at different distances from the shore.

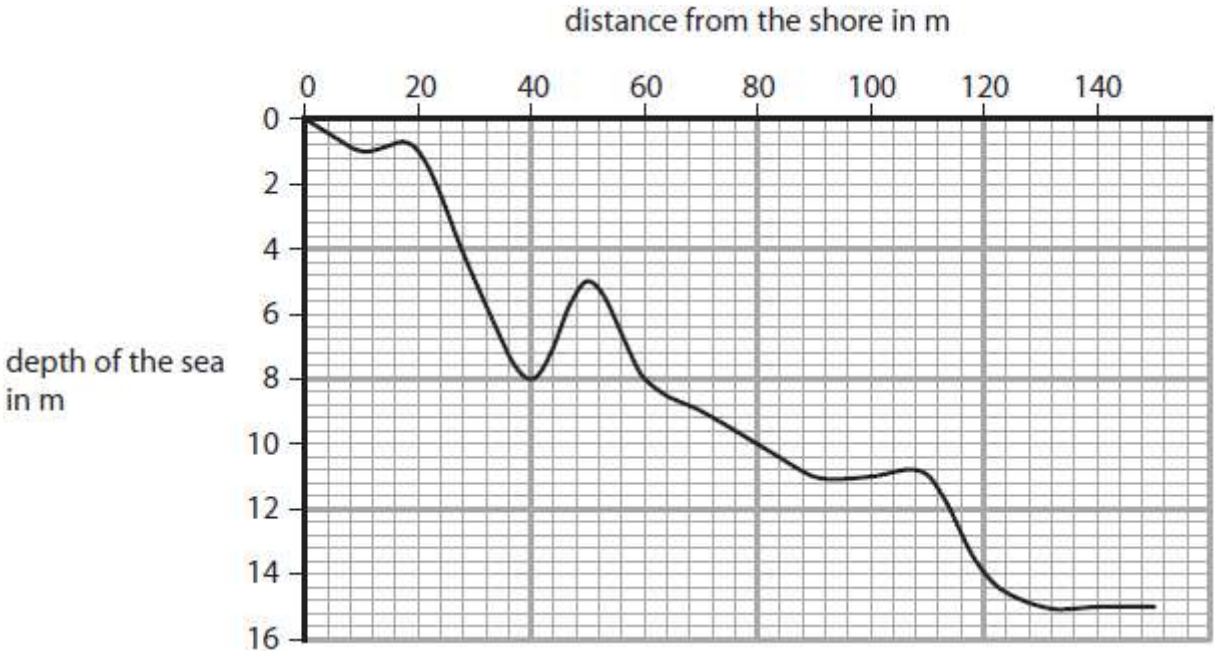

Figure 1

A technician on a boat uses sonar pulses to measure the depth of the sea when the boat is 120 m from the shore.

Calculate the **total** time of travel for the sonar pulse used to make this measurement.

The speed of the sonar pulse in seawater is 1600 m/s.

(4)

time of travel = ... s

(Total for question = 4 marks)

Q3.

(a) A man uses a dog whistle to call his dog.
The whistle uses ultrasound.

(i) The dog can hear the whistle but the man cannot.
Explain why the dog can hear the whistle but the man cannot hear the whistle.

(2)

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(ii) The dog is 140 m away from the man.
The ultrasound takes 0.42 s to travel from the man to the dog.
Calculate the speed of ultrasound.
State the unit.

(3)

speed of ultrasound =..... unit = ...

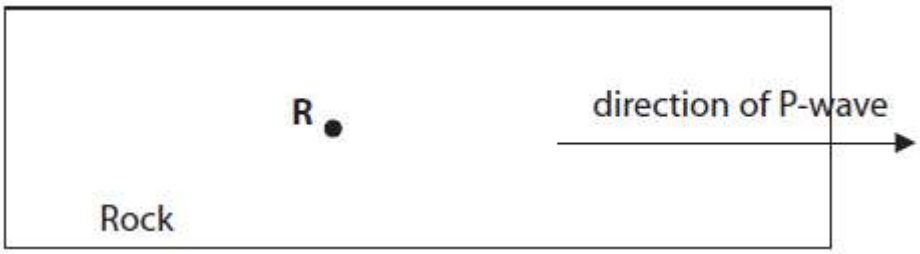
(b) (i) An earthquake P-wave has a frequency of 15 Hz.

Complete the sentence by putting a cross (☒) in the box next to your answer.
The earthquake P-wave is

(1)

☒ **A** an infrasound wave
☒ **B** an ultrasound wave
☒ **C** an electromagnetic wave
☒ **D** a transverse wave

(ii) The diagram shows the direction of an earthquake P-wave in some rock.



Draw arrows on the diagram to show how the piece of rock, labelled **R**, moves.

(2)

(c) Earthquakes occur when two tectonic plates move against each other.
Explain what causes the tectonic plates to move.

You may draw a diagram if it helps your answer.

(2)

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(Total for Question = 10 marks)

Q4.

The photograph shows a meteor exploding above Russia in 2013.

(a) The noise from the explosion was described as the loudest sound ever detected on Earth.
However, human beings could not hear this sound.

State the **two** sound frequency ranges that human beings cannot hear.

(2)

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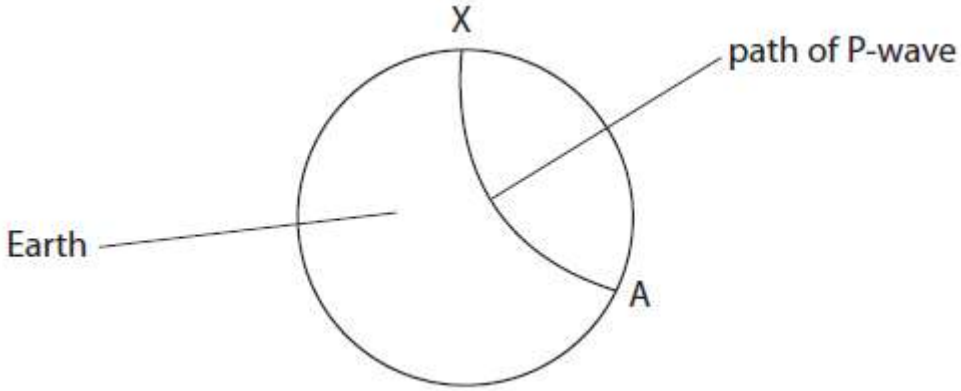
(b) Meteors sometimes collide with the Earth's surface.
These collisions produce both P-waves and S-waves.

(i) Which of the following is correct for a P-wave?
Put a cross (☒) in the box next to your answer.

(1)

- ☐ **A** It is a transverse wave travelling faster than an S-wave.
- ☐ **B** It is a transverse wave travelling slower than an S-wave.
- ☐ **C** It is a longitudinal wave travelling faster than an S-wave.
- ☐ **D** It is a longitudinal wave travelling slower than an S-wave.

The diagram shows the path of a P-wave.
The P-wave travels from the collision at X, through the Earth, to another point, A.



(ii) Explain why the path of the P-wave is not a straight line.

(2)

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(iii) Explain why there are regions on the Earth's surface where S-waves from the collision at X cannot be detected. You can add to the diagram to help your answer.

(3)

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(iv) Describe how a meteor colliding with the Earth could set off an earthquake.

(2)

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(Total for Question = 10 marks)

Q5.

Light waves and sound waves are both used in the diagnosis and treatment of medical conditions

(i) A doctor uses an endoscope to look inside the body of a patient.
Explain how optical fibres are used in endoscopes.
You may draw a labelled diagram to help with your answer.

(3)

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(ii) Describe how ultrasound can be used as a medical treatment for illness or injury.

(2)

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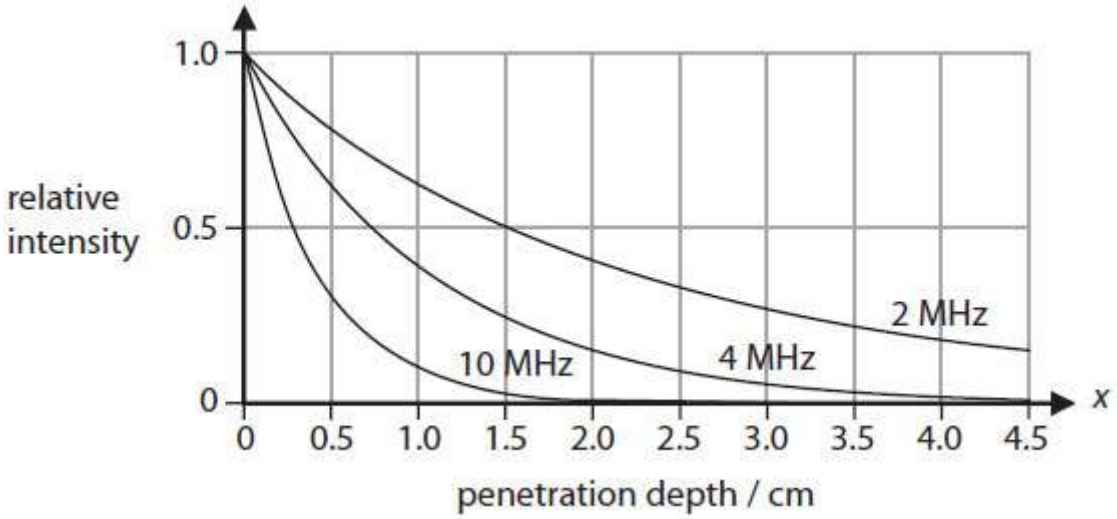
Q6.

The graph shows how the intensity of ultrasound waves of different frequencies decreases as they penetrate soft tissue.



(i) Estimate how far a 2 MHz wave has penetrated into the soft tissue when its intensity is 25% of its original value.

(1)

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(ii) Explain which of these frequencies of ultrasound can be used to scan organs deep inside the body.

(2)

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Q7.

Many earthquakes and volcanoes are linked to the production of infrasound waves.
Describe what is meant by **infrasound waves.**

(2)

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Q8.

When an earthquake happens, waves are transmitted through the Earth.
Figure 6 is a diagram of a part of the Earth's crust.

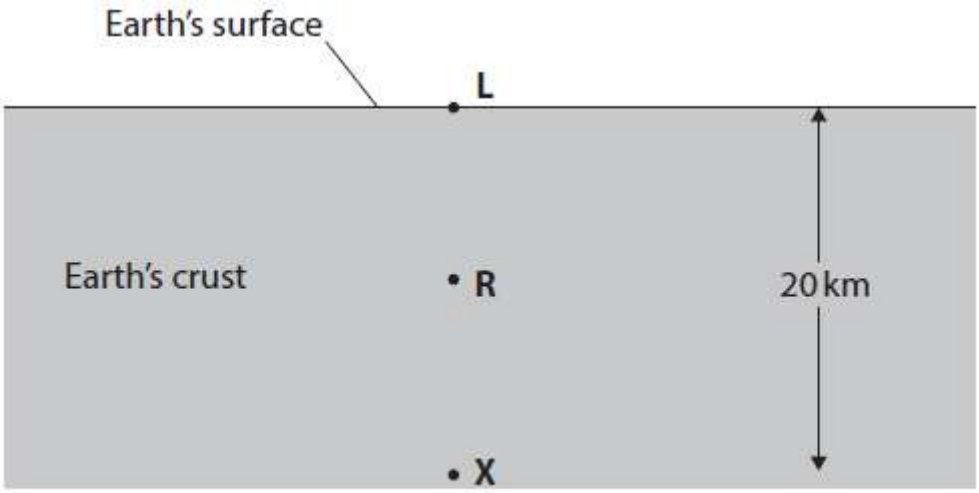

Figure 6

An earthquake happens at point **X**, 20 km below the Earth's surface.

An earthquake wave from **X** is detected at point **L**.

This earthquake wave is a transverse wave.

(i) State **one** other example of a transverse wave.

(1)

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(ii) A point in the crust is labelled **R** in Figure 6.

There is a rock at point **R**.
Describe the effect that this earthquake wave has on the rock at **R** as the wave passes through the rock.
You may add to Figure 6 to help your answer.

(2)

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(iii) The wave from **X** reaches **L** 3.8 seconds after it passes through **R**.

The speed of the wave is 4500 m/s.
Calculate the distance between **X** and **R**.

(2)

distance = ... m

(Total for question = 5 marks)

Q9.

Figure 9 is a table of data for three coloured beams of light travelling from air into glass.

	wavelength in air	frequency in air	wavelength in glass	frequency in glass
red light	650 nm	4.6×10^{14} Hz	490 nm	4.6×10^{14} Hz
yellow light	580 nm	5.2×10^{14} Hz	440 nm	5.2×10^{14} Hz
blue light	470 nm	6.4×10^{14} Hz	350 nm	6.4×10^{14} Hz

Figure 9

Using data from Figure 9,

(i) state what happens to the wavelength of light when the light travels from glass into air

(1)

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(ii) state what happens to the frequency of light when the light travels from glass into air.

(1)

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(iii) The speed of blue light in air is 3.0×10^8 m/s.

Calculate the change in the speed of blue light when it travels from air into glass.
Use the equation

$$v = f \times \lambda$$

(3)

change in speed = ... m/s

(Total for question = 5 marks)

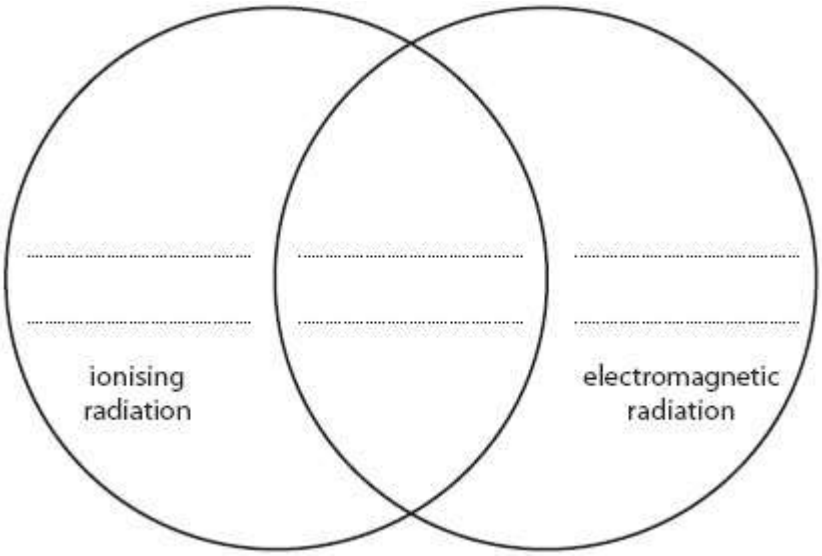
Q10.

(a) Skin cancer can be caused by radiation from the Sun.

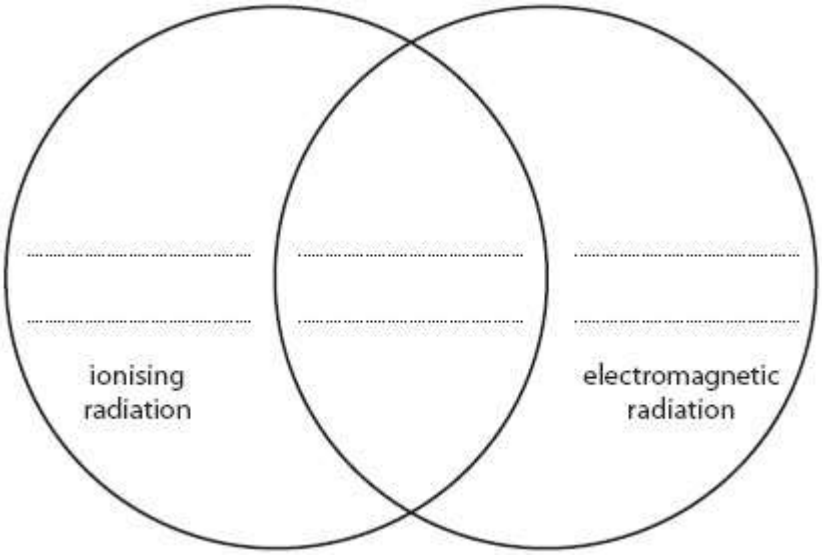
Complete the sentence by putting a cross (■) in the box next to your answer.

The radiation that causes skin cancer is

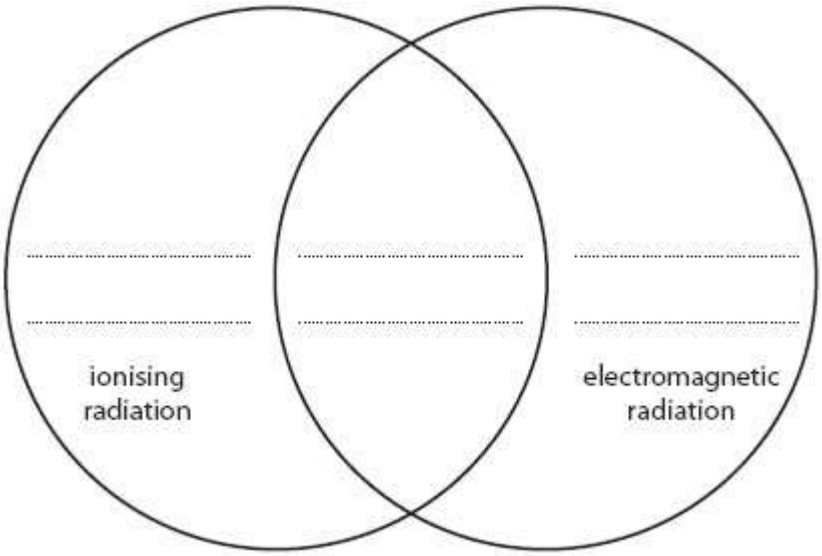
(1)



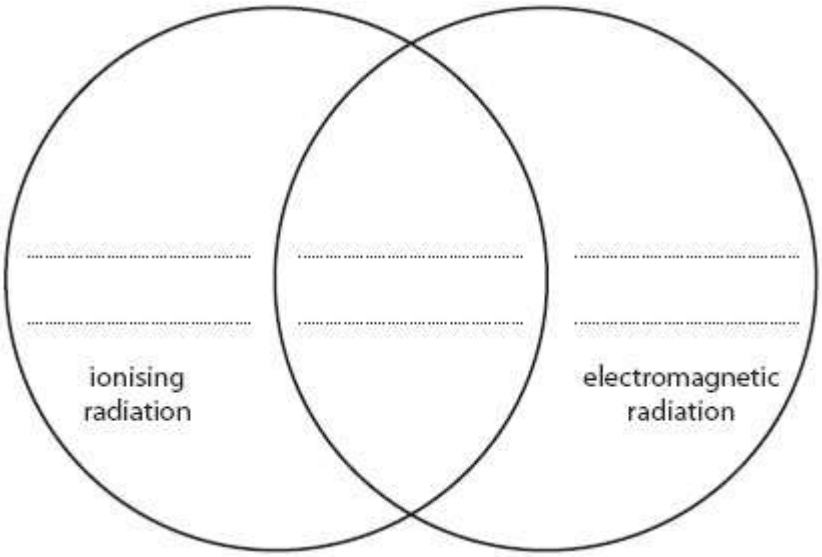
A ultraviolet radiation



B radio waves



C microwaves



D infrared radiation

(b) The word box contains the names of three types of radiation.

gamma rays	infrared radiation	alpha particles

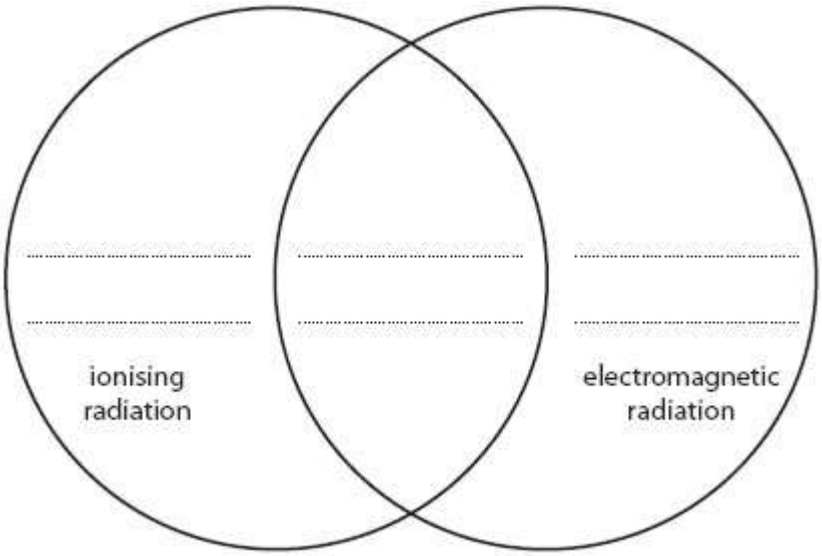
Use this diagram to classify the three types of radiation given in the word box.
Write the name of the radiation in the correct section of the diagram.

(2)

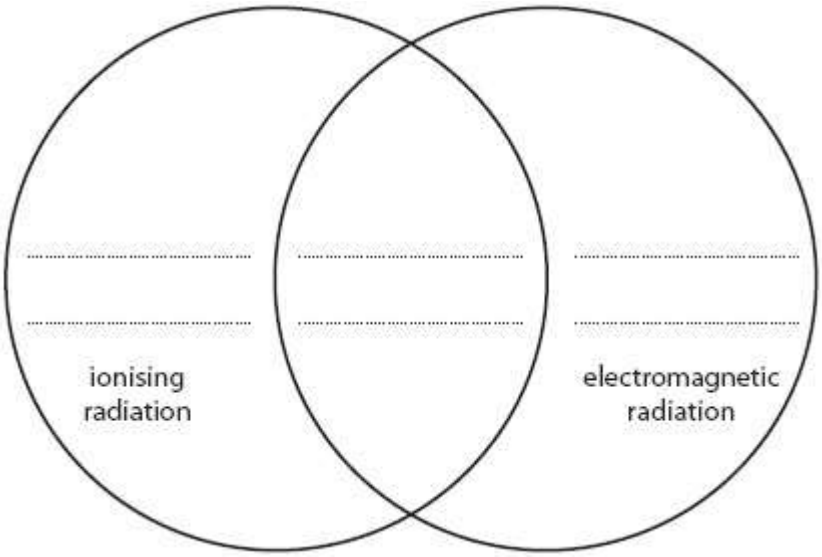
(c) Which of these is correct for all electromagnetic waves in a vacuum?

Put a cross (■) in the box next to your answer.

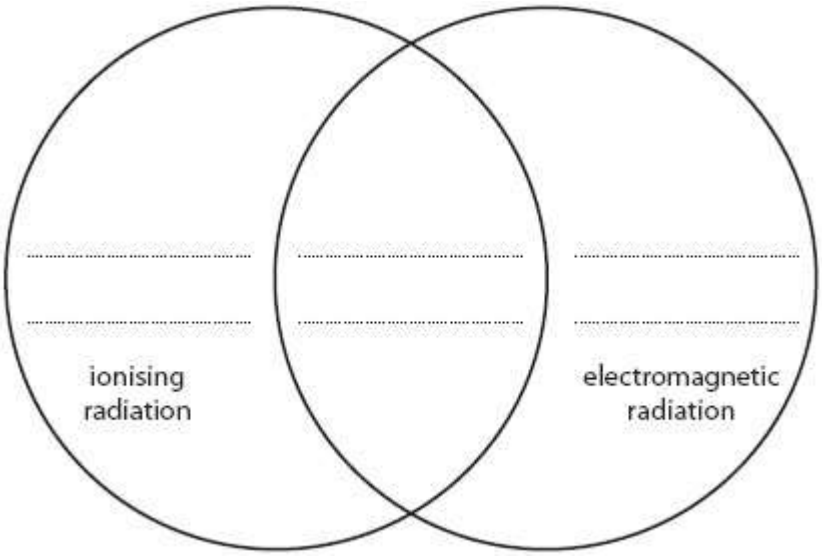
(1)



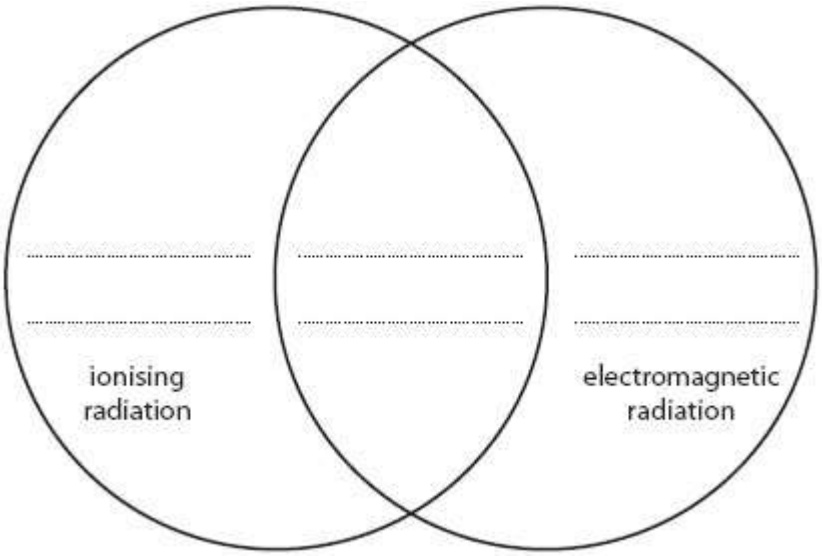
A they have the same frequency



B they have the same wavelength



C they are transverse waves



D they are longitudinal waves

(d) Describe a use of gamma radiation.

(2)

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*(e) Herschel and Ritter carried out experiments that contributed to the discovery of infrared and ultraviolet radiation.

Explain how the results of the experiments carried out by Herschel and Ritter led to these discoveries.

(6)

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(Total for Question is 12 marks)

Mark Scheme

Q1.

Question Number	Answer	Additional guidance	Mark
	an explanation linking: (the colours have) different wavelengths (1) different wavelengths / colours travel at different speeds (1) so refract by different amounts (1)	 allow the word frequencies for wavelengths for refract allow bend/change direction/follow different path	**(3)** AO 2 1

Q2.

Question Number	Answer	Additional guidance	Mark
	recall (1) $v = \frac{x}{t}$ rearrangement (1) $t = \frac{x}{v}$ substitution (1) $\frac{14 \times 2}{1600}$ evaluation (1) 0.018 (s)	 substitution and rearrangement in either order max 3 marks if 14 used as distance accept numbers that round to 0.018 e.g. 0.0175 (s) award full marks for the correct answer with no working	**(4)** AO 1 1 AO 2 1

Q3.

Question Number	Answer	Acceptable answers	Mark
(a)(i)	an explanation linking: • frequency / Hz (1) • above 20 000 (1)	Pitch too high to be heard by the man "it is above 20kHz" 2 marks "The frequency is too loud" gets 1st mark	**(2)**

Question Number	Answer	Acceptable answers	Mark
(a)(ii)	substitution: (1) 140/0.42 evaluation: (1) 330 m/s (1)	award full marks for correct answer with no working allow 333(.333) independent mark allow ms^{-1}	**(3)**

Question Number	Answer	Acceptable answers	Mark
(b)(i)	A infrasound wave (1)		**(1)**

Question Number	Answer	Acceptable answers	Mark
(b)(ii)	• arrows to show vibration in opposite directions (1) • parallel to arrow on diagram (1)	arrows do not have to go through R horizontal and vertical – no marks multiple directions – no marks	**(2)**

Question Number	Answer	Acceptable answers	Mark
(c)	Explanation linking: • convection (currents) (1) • in mantle (1)	Accept answers in form of a labelled diagram in molten rock in magma below plates in the hot rock coming from the core under Earth's crust under surface ignore lava clear unlabelled diagram scores maximum 1 mark clear labelled diagram scores maximum 2 marks	**(2)**

Q4.

Question Number	Answer	Acceptable answers	Mark
(a)	• below 20 Hz (1) • above {20 000 Hz / 20 kHz} (1) If Hz or kHz is not seen somewhere, the maximum score is 1 mark.	infrasound ultrasound (in either order) (no units needed for the names)	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b)(i)	C it is a longitudinal wave travelling faster than an S wave		**(1)**

Question Number	Answer	Acceptable answers	Mark
(b)(ii)	Explanation linking the following:- MP1 refraction /changing speed (1) MP2 (due to) changing {material/medium /rock type / density} (1)	ignore changes in direction/ bending (in this case) rock becomes {more / less} {dense / compact}	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b)(iii)	Explanation linking the following:- MP1 (S / transverse waves) they cannot travel through liquid (1) MP2 Earth's core is (at least part) {liquid/molten} (1) MP3 (so) (S waves) they cannot travel through core (to other side of Earth) (1)	Check diagram for creditworthy points. they can only travel through solids may be stated in part (ii) (S / transverse waves) they cannot be detected on opposite side of the Earth to (collision site / earthquake)	**(3)**

Question Number	Answer	Acceptable answers	Mark
(b)(iv)	Suggestion to include any two from: MP1 idea that {kinetic energy/force/ momentum} of meteor might cause the earthquake (1) MP2 (earthquakes happen where) plates slide {past/over/under/away from/against} each other (1) MP3 (plates move) suddenly MP4 (meteor collision) starts seismic waves /P/S (1)	(meteor) it has large amount of kinetic energy (earthquakes happen where) plates collide rub/move for slide (earthquakes happen when) large amount of energy released in / near Earth's surface (plates) jolt/jerk vibrations passing through the Earth condone earthquake waves {kinetic energy/force /momentum} of meteor can cause the plates to slide past each other = 2	**(2)**

Q5.

	Answer	**Acceptable answers**	**Mark**
(i)	An explanation linking any three of the following: (Optical fibres) bend (1) some fibres carry	Accept suitable labelling on a Diagram Image projected on a screen	**(3)**

	light to the inside of the patient (1) some fibres transmit the reflected light (1) light passes up/down fibres by TIR (1) light is reflected inside the patient (1) image is analysed by computer (1)		
(ii)	Either Breaks/blasts/smashes (1) Kidney stones (1) or Energy absorbed (1) to help repair muscle tissue (1) or Use of gel (1) to prevent loss of intensity (1) or	bruising/clots increases blood flow Allow (1) mark for suitable diagnosis e.g. prenatal scan	**(2)**

Q6.

	Answer	**Acceptable answers**	**Mark**
(i)	3.0 +/- 0.5 (cm)		**(1)**
(ii)	an explanation linking • 2 MHz (1)and any **one** from: • has a higher intensity inside tissue (1) • less energy absorbed (1) • less attenuation (1) • penetrates furthest /deepest (1)	this frequency alone RA loses intensity more gradually highest penetration accept "2MHz and 4MHz" with correct reason for 1 mark	**(2)**

Q7.

	Answer	**Acceptable answers**	**Mark**
	a description including **two** of the following: • longitudinal /	Frequency below range/too low for (normal) human ear	**(2)**

	sound (wave) (1) • (frequency) less than 20 Hz (1)		

Q8.

Question Number	Answer	Additional guidance	Mark
(i)	any **one** from electromagnetic (wave) (1) ripples/water (waves) (1) (waves on a) string (1)	 radio(waves) / microwaves / IR / infrared / visible / light / UV / ultraviolet / X(-ray) / gamma / γ allow sea / ocean / tidal wave ignore waves on a spring ignore earthquake or seismic P or S	**(1)** **AO1.1**

Question Number	Answer	Additional guidance	Mark
(ii)	a description to include: (causes it to) vibrate/oscillate (1) perpendicular/at right angles to (the direction of travel of) the wave (1)	can be shown on diagram allow shake in this context allow double headed arrow(s) ↔ or ↕ centred on R on diagram (to show vibration) allow "up and down" or "forwards and backwards" for vibration perpendicular/at right angles to (direction of) energy transfer allow horizontal arrow(s) → or ← centred on R on diagram (to show direction) allow side to side OR left to right OR in and out for perpendicular ↔ centred on R scores 2 marks ↔ centred on R scores 2 marks ignore possible eventual consequences e.g. rock breaks up / heats up	(2) AO2.1

Question Number	Answer	Additional guidance	Mark
(iii)	calculation of distance from R to L (1) (= 4500 x 3.8) = 17 000 (m) calculation of distance from X to R (1) (= 20 000 – 17 000) = 3000 (m) ALTERNATIVE METHOD calculation of time X to L (1) (20 000 / 4500 =) 4.44 calculation of distance from X to R (1) (=(4.44 - 3.8) x 4500) = 3000 (m)	 accept 17 100 allow ECF from MP1 accept numbers which round to 3000 e.g. 2900 accept numbers which round to 3000 e.g. 2900 award full marks for the correct answer without working	(2) AO2.1

Q9.

	Answer	Additional guidance	Mark
(i)	(wavelength) increases (from glass to air)	gets longer/gets bigger allow quoted values allow (wavelength) decreases from air to glass	(1) AO3.1

	Answer	Additional guidance	Mark
(ii)	(frequency) doesn't change	remains constant / stays the same / nothing allow quoted values	(1) AO3.1

	Answer	Additional guidance	Mark
(iii)	substitution (1) (speed in glass =) 6.4 (× 10^{14}) × 350 (× 10^{-9}) evaluation of speed in glass (1) 2.2 × 10^{8} (m/s) subtraction involving 3 × 10^{8} (m/s) (1) ALTERNATIVE METHOD calculation of change in wavelength for blue light (1) 470(× 10^{-9}) - 350 (× 10^{-9}) OR 120 (× 10^{-9}) substitution (1) (difference in speed =) 6.4 (× 10^{14}) x 120 (× 10^{-9}) substitution (1) (difference in speed =) 6.4 (× 10^{14}) x 120 (× 10^{-9}) evaluation (1) 7.7 × 10^{7} (m/s)	allow numbers that round to 2.2 × 10^{8} e.g. 2.24 × 10^{8} independent mark 7.6 × 10^{7} (m/s) scores 3 marks ecf their change in wavelength for this mark only ecf their change in wavelength for this mark only award full marks for the correct answer without working	**(3)** **AO3.2**

Q10.

	Answer	Acceptable answers	Mark
(a)	A		**(1)**
(b)	alpha particles (In the left section) gamma rays (centre section) infrared radiation (right section) (2)	Any one in correct position for one mark, all three in correct position for two marks	**(2)**
(c)	C		**(1)**
(d)	A description to include The purpose of using gamma radiation (1) Some relevant detail about how it achieves the purpose (1)	Purposes may include sterilising food /medical equipment detection / treatment of cancer imaging /detect flaws in materials	**(2)**

		Indicative Content	Mark
QWC	***(e)**	An explanation including some of the following points Results obtained: • Herschel: temperature on thermometer • Ritter: speed of darkening of silver chloride paper Trend of results: • Herschel: hotter towards red end • Ritter: quicker towards blue/violet end Extension of experiment to get more results: • Herschel: measure below red; found it even hotter • Ritter: measure above blue/violet; paper darkened quicker Conclusion: • Herschel: Must be radiation below red (Infra Red) • Ritter: Must be radiation above blue/violet (UV)	**(6)**
Level	**0**	No rewardable content	
1	**1 - 2**	• A limited description of either some results or conclusions from either	

		experiment. For example: They measured temperature across the spectrum and found that temperature changed. They put silver chloride paper in the spectrum and found that it darkened at different speeds with different colours. The answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy
2	**3 - 4**	• A simple explanation of results and conclusions from both experiments . For example: Herschel measured the temperature across the spectrum and found it hotter towards the red end. This was infra red radiation. Ritter measured the darkening of chloride paper across the spectrum. It was quicker towards the violet end. They had discovered ultra violet. • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy
3	**5 - 6**	• a detailed explanation of all the results obtained from both experiments and the conclusions from these results. For example a response as for level 2 given above but with detail about results being obtained from outside the visible spectrum • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors